

A Block-Measure Criterion for Copies of c_0 in Adequate-Family Spaces

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Status. Full answer, likely valid. The source paper asks for a condition on an adequate family $\mathcal{A}$ guaranteeing that $h_{\mathcal{A},p}$ does not contain c_0 , equivalently that its canonical basis is boundedly complete. The answer below gives an exact block-measure criterion.

Source Question

The source is Trond A. Abrahamsen, Vegard Lima, and Andre Martiny, *Delta-points in Banach spaces generated by adequate families*, arXiv:2012.00406. In the final section, after proving their main results on delta-points and polyhedrality, the authors write:

“Note that $h_{\mathcal{A},1}$ and even $h_{\mathcal{A},p}$ contains an isometric copy of c_0 if there exists an infinite subset $E \subseteq \mathbb{N}$ such that $|A \cap E| \leq 1$ for all $A \in \mathcal{A}$. We ask:

What is a natural condition on $\mathcal{A}$ such that $h_{\mathcal{A},1}$ (or $h_{\mathcal{A},p}$) does not contain c_0 . That is, when does $h_{\mathcal{A},p}$ have a boundedly complete basis?”

is infinite.

□

We end the paper with some questions.

Let $\mathcal{A}$ be an adequate family. Proposition 4.3 gives a simple characterization of when $h_{\mathcal{A},1}$ contains a copy of ℓ_1 and $h_{\mathcal{A},p}$ never contains ℓ_1 by Lemma 3.2 (and James’ characterization of shrinking unconditional bases).

Note that $h_{\mathcal{A},1}$ and even $h_{\mathcal{A},p}$ contains an isometric copy of c_0 if there exists an infinite subset $E \subseteq \mathbb{N}$ such that $|A \cap E| \leq 1$ for all $A \in \mathcal{A}$. We ask:

Question 5.5. What is a natural condition on $\mathcal{A}$ such that $h_{\mathcal{A},1}$ (or $h_{\mathcal{A},p}$) does not contain c_0 . That is, when does $h_{\mathcal{A},p}$ have a boundedly complete basis?

We have seen that if $\mathcal{A}$ is an adequate family of finite sets which is spreading, then neither $h_{\mathcal{A},1}$ nor $h_{\mathcal{A},1}^*$ have delta-points. We do not know if that is also the case if $\mathcal{A}$ is not spreading.

Question 5.6. If $h_{\mathcal{A},1}$ is (V)-polyhedral, does then $h_{\mathcal{A},1}^*$ fail to have delta-points?

Definitions

An adequate family $\mathcal{A}$ of subsets of $\mathbb{N}$ contains the empty set and all singletons, is hereditary, and is compact in the topology of pointwise convergence. For $1 \leq p < \infty$, $h_{\mathcal{A},p}$ is the completion of c_{00} under the norm

$$\left\| \sum_i a_i e_i \right\| = \sup_{A \in \mathcal{A}} \left(\sum_{i \in A} |a_i|^p \right)^{1/p}.$$

The canonical basis is normalized and 1-unconditional.

For a finitely supported nonnegative measure μ on $\mathbb{N}$, write

$$\|\mu\|_{\mathcal{A}} = \sup_{A \in \mathcal{A}} \mu(A).$$

A sequence (μ_n) of finitely supported nonnegative measures is *successive* if their supports are finite intervals in increasing order, or more generally finite sets $I_n = \text{supp } \mu_n$ satisfying $\max I_n < \min I_{n+1}$. It is an $\mathcal{A}$ -bounded normalized block-measure sequence if

$$\|\mu_n\|_{\mathcal{A}} = 1 \quad (n \in \mathbb{N})$$

and

$$C(\mu) := \sup_{m \in \mathbb{N}} \sup_{A \in \mathcal{A}} \sum_{n=1}^m \mu_n(A) < \infty.$$

Idea of the Proof

The norm of $h_{\mathcal{A},p}$ reads the p -mass of a vector through the sets in $\mathcal{A}$. Thus a normalized positive block x_n determines a normalized measure $\mu_n(i) = |x_n(i)|^p$, and the norm of a positive sum of such blocks is exactly the p -th root of the accumulated $\mathcal{A}$ -mass of those measures.

This makes the c_0 obstruction transparent. If the accumulated $\mathcal{A}$ -mass of normalized block measures is uniformly bounded, the associated blocks have uniformly bounded finite sums and are therefore equivalent to the unit vector basis of c_0 . Conversely, any copy of c_0 in a space with a Schauder basis can be gliding-humped into a block sequence equivalent to the c_0 basis. Taking coordinatewise absolute values and p -th powers of that block sequence gives the required bounded block-measure sequence.

Theorem 1 (Block-measure criterion). *Let $\mathcal{A}$ be an adequate family and let $1 \leq p < \infty$. Then $h_{\mathcal{A},p}$ contains an isomorphic copy of c_0 if and only if there is an $\mathcal{A}$ -bounded normalized block-measure sequence.*

Equivalently, $h_{\mathcal{A},p}$ contains no copy of c_0 if and only if every successive sequence (μ_n) of finitely supported nonnegative measures with $\|\mu_n\|_{\mathcal{A}} = 1$ satisfies

$$\sup_{m \in \mathbb{N}} \sup_{A \in \mathcal{A}} \sum_{n=1}^m \mu_n(A) = \infty.$$

Proof. First assume that (μ_n) is an $\mathcal{A}$ -bounded normalized block-measure sequence. Let $I_n = \text{supp } \mu_n$, and define

$$x_n = \sum_{i \in I_n} \mu_n(\{i\})^{1/p} e_i.$$

Then

$$\|x_n\|^p = \sup_{A \in \mathcal{A}} \sum_{i \in A} \mu_n(\{i\}) = \|\mu_n\|_{\mathcal{A}} = 1.$$

Let $a_1, \dots, a_m$ be scalars and put

$$M = \max_{1 \leq n \leq m} |a_n|.$$

Since the canonical basis is 1-unconditional, coordinate projections onto arbitrary sets have norm at most one. Hence, for each $k \leq m$,

$$|a_k| = \|a_k x_k\| = \left\| P_{I_k} \sum_{n=1}^m a_n x_n \right\| \leq \left\| \sum_{n=1}^m a_n x_n \right\|.$$

Taking the maximum over k gives the lower c_0 estimate.

For the upper estimate, by the definition of the norm,

$$\left\| \sum_{n=1}^m a_n x_n \right\|^p = \sup_{A \in \mathcal{A}} \sum_{n=1}^m |a_n|^p \mu_n(A) \leq M^p \sup_{A \in \mathcal{A}} \sum_{n=1}^m \mu_n(A) \leq M^p C(\mu).$$

Thus (x_n) is equivalent to the unit vector basis of c_0 , and $h_{\mathcal{A},p}$ contains c_0 .

Conversely suppose that $h_{\mathcal{A},p}$ contains a subspace isomorphic to c_0 . Let (u_n) be the unit vector basis of c_0 and let $T : c_0 \rightarrow h_{\mathcal{A},p}$ be an isomorphic embedding. Then Tu_n is a seminormalized weakly null basic sequence equivalent to the c_0 basis. Since $h_{\mathcal{A},p}$ has a Schauder basis, the standard gliding-hump form of the Bessaga-Pelczynski selection principle gives a subsequence and a normalized block basic sequence (z_n) , with successive finite supports I_n , which is equivalent to a subsequence of (Tu_n) . In particular, there is $K < \infty$ such that

$$\left\| \sum_{n=1}^m b_n z_n \right\| \leq K \max_{1 \leq n \leq m} |b_n|$$

for all finite scalar choices (b_n) .

Because the canonical basis of $h_{\mathcal{A},p}$ is 1-unconditional and the supports of the z_n 's are disjoint,

$$\left\| \sum_{n=1}^m |z_n| \right\| = \left\| \sum_{n=1}^m z_n \right\| \leq K,$$

where $|z_n|$ denotes the coordinatewise absolute value. Define

$$\mu_n(\{i\}) = |z_n(i)|^p \quad (i \in I_n), \quad \mu_n(\{i\}) = 0 \quad (i \notin I_n).$$

Then

$$\|\mu_n\|_{\mathcal{A}} = \sup_{A \in \mathcal{A}} \sum_{i \in A} |z_n(i)|^p = \|z_n\|^p = 1.$$

Moreover, for every m ,

$$\sup_{A \in \mathcal{A}} \sum_{n=1}^m \mu_n(A) = \left\| \sum_{n=1}^m |z_n| \right\|^p \leq K^p.$$

Thus (μ_n) is an $\mathcal{A}$ -bounded normalized block-measure sequence. □

Corollary 1 (Answer to the boundedly-complete-basis question). *Let $\mathcal{A}$ be an adequate family and $1 \leq p < \infty$. The canonical basis of $h_{\mathcal{A},p}$ is boundedly complete if and only if $\mathcal{A}$ admits no $\mathcal{A}$ -bounded normalized block-measure sequence.*

Proof. By James' classical theorem for unconditional bases, a Banach space with an unconditional basis contains no copy of c_0 if and only if that basis is boundedly complete. Apply this to the normalized 1-unconditional basis of $h_{\mathcal{A},p}$ and use Theorem 1. $\square$

Corollary 2 (The source sufficient condition). *If there is an infinite set $E = \{i_1 < i_2 < \dots\} \subseteq \mathbb{N}$ and a constant $M < \infty$ such that*

$$|A \cap E| \leq M \quad (A \in \mathcal{A}),$$

then $h_{\mathcal{A},p}$ contains an isomorphic copy of c_0 . In the case $M = 1$ this is the isometric-copy observation recorded in the source paper.

Proof. Take $\mu_n = \delta_{i_n}$. Then $\|\mu_n\|_{\mathcal{A}} = 1$ because singletons belong to $\mathcal{A}$, and

$$\sup_m \sup_{A \in \mathcal{A}} \sum_{n=1}^m \mu_n(A) = \sup_m \sup_{A \in \mathcal{A}} |A \cap \{i_1, \dots, i_m\}| \leq M.$$

Theorem 1 applies. $\square$

Verification and Novelty Check

The proof is a direct norm computation plus two standard facts about Schauder bases: the gliding-hump/block selection principle for seminormalized weakly null sequences, and James' bounded-completeness characterization for unconditional bases. The source paper itself records that the canonical basis of $h_{\mathcal{A},p}$ is normalized and 1-unconditional.

Local run indexes were searched for arXiv:2012.00406 and the phrases *adequate family*, *delta-points in Banach spaces generated by adequate families*, $h_{\mathcal{A},p}$, c_0 , and *boundedly complete basis*. No existing packet for this question was found. Web searches on June 20, 2026 for the exact title, the exact polyhedral/delta-point question, $h_{\mathcal{A}}^*$ *delta-points adequate family*, and close variants found the source paper but no later paper giving this block-measure characterization.

Limitations

The theorem answers the first closing question in the source paper, namely the c_0 /boundedly-complete-basis question. It does not answer the separate question asking whether (V)- or (I)-polyhedrality of $h_{\mathcal{A},1}$ forces $h_{\mathcal{A},1}^*$ to fail to have delta-points.

Human Review Recommendation

Review should focus on the converse implication, specifically the use of the standard fact that a copy of c_0 in a Banach space with a Schauder basis has a block basic sequence equivalent to the c_0 basis. After that selection step, the rest of the argument is an explicit calculation in the defining norm of $h_{\mathcal{A},p}$.

References

- [1] T. A. Abrahamsen, V. Lima, and A. Martiny, *Delta-points in Banach spaces generated by adequate families*, arXiv:2012.00406, 2020.
- [2] C. Bessaga and A. Pelczynski, *On bases and unconditional convergence of series in Banach spaces*, Studia Math. 17 (1958), 151–164.
- [3] R. C. James, *Bases and reflexivity of Banach spaces*, Ann. of Math. 52 (1950), 518–527.